

Fast algorithms to efficiently capture the adaptive concentration for dispersal evolution model

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Abstract

The evolution of a dispersal trait is a classical question in evolutionary ecology. A typical phenomenon is that the trait variable will concentrate to a Dirac mass, which makes the accurate capture of the fittest trait difficult. In our paper, we presented a new scheme based on WKB ansatz to efficiently capture the fittest trait without putting many points in the phenotypic direction. The boundedness of the density is proven. Numerical examples are provided to show the efficiency of our scheme.

Key words.

1 Introduction

Trait-structured population models have become a standard mathematical framework to describe how ecological and evolutionary processes interact across multiple scales. In particular, dispersal evolution is a central topic in evolutionary ecology: a population distributed in space may carry a heritable dispersal trait, and natural selection may favor specific dispersal strategies depending on the spatial heterogeneity of resources and competition [?, ?]. A prototypical model, already widely used in the literature, takes the form

$$\varepsilon \partial_t n_\varepsilon - D(\theta) \Delta_{\mathbf{x}} n_\varepsilon - \varepsilon^2 \Delta_\theta n_\varepsilon = n_\varepsilon (K(\mathbf{x}) - \rho_\varepsilon(\mathbf{x})), \quad \mathbf{x} \in \Omega, \quad (1.1)$$

where t denotes time, $\mathbf{x}$ the spatial variable and θ a phenotypic (trait) variable. The unknown $n_\varepsilon(t, \mathbf{x}, \theta)$ is the structured population density, and

$$\rho_\varepsilon(t, \mathbf{x}) = \int_{\mathbb{T}} n_\varepsilon(t, \mathbf{x}, \theta) d\theta$$

is the total population density at $(t, \mathbf{x})$. The function $K(\mathbf{x})$ represents the local carrying capacity, and the trait-dependent diffusivity $D(\theta)$ models the dispersal rate associated with phenotype θ . Throughout this work we impose homogeneous Neumann boundary conditions on $\partial\Omega$ and periodic boundary conditions in the θ -direction.

Model (1.1) (and its variants) has been extensively investigated from the analytical viewpoint, including well-posedness, long-time behavior, and the connection with adaptive dynamics. A key feature is that dispersal evolution is encoded through the coupling between spatial heterogeneity (via $K(\mathbf{x})$) and trait-dependent dispersal (via $D(\theta)$), which may select an “optimal” phenotype in

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the long-time regime [?, ?, ?]. In many situations, solutions converge as $t \rightarrow +\infty$ to a steady state, and the trait distribution exhibits concentration phenomena.

The small parameter $\varepsilon > 0$ plays the role of a rare mutation scale: the trait diffusion is of order ε^2 , while time is rescaled by ε . In the rare mutation limit, the steady state typically concentrates in the trait variable and converges (in a weak sense) to a Dirac mass supported at the fittest trait. This singular behavior is now well-understood at the PDE level via WKB/Hopf–Cole transforms and constrained Hamilton–Jacobi limits [?, ?, ?]. From the numerical perspective, however, this regime is notoriously challenging: as $\varepsilon \rightarrow 0$, n_ε develops extremely sharp peaks in θ , so a direct discretization of (1.1) requires a prohibitively fine mesh in the phenotypic direction in order to locate the fittest trait accurately. In addition, the coupling through the nonlocal density ρ_ε and the trait-dependent diffusion $D(\theta)$ further complicate stability analysis and the design of efficient schemes.

Motivated by the asymptotic structure of the rare mutation limit, we follow a classical WKB strategy and rewrite the solution as

$$n_\varepsilon(\mathbf{x}, \theta, t) = W_\varepsilon(\mathbf{x}, \theta, t) \exp\left(\frac{u_\varepsilon(\theta, t)}{\varepsilon}\right),$$

where u_ε captures the rapidly varying exponential profile in the trait variable, while W_ε is a smoother amplitude. Formally, in the long-time and rare-mutation regime, the effective dynamics of the phase function is governed by a constrained Hamilton–Jacobi equation involving an effective Hamiltonian $\mathcal{H}(\theta, \rho)$ defined through a principal eigenvalue problem in the spatial variable [?, ?]. This reformulation suggests that one can design an *asymptotic-preserving* (AP) scheme by evolving $(u_\varepsilon, W_\varepsilon)$ while consistently incorporating the effective Hamiltonian. The potential benefit is substantial: the location of the fittest trait can be extracted from u_ε without resolving the narrow peak of n_ε on a very fine θ -grid.

The goal of this paper is to develop and analyze fast numerical algorithms that efficiently capture the adaptive concentration in the trait variable. Our contributions can be summarized as follows:

- We propose a WKB-based reformulation and a semi-discrete numerical scheme that couples (i) a monotone approximation of the Hamilton–Jacobi equation for u_ε and (ii) a positivity-preserving discretization for the amplitude W_ε , together with a consistent evaluation of the effective Hamiltonian through a discrete principal eigenvalue problem.
- We establish key *a priori* properties of the scheme, including positivity and uniform-in- ε boundedness of the discrete density ρ^ε . The boundedness proof relies on a weighted argument obtained by dividing the discrete n -equation by $D(\theta)$ before summation in θ , which avoids spurious mesh-dependent constants and highlights the role of the regularity of $1/D$.
- We provide numerical experiments demonstrating that the proposed method can capture the fittest trait accurately using far fewer grid points in the phenotypic direction than direct discretizations of (1.1), especially in the small- ε regime where Dirac concentration occurs.

The remainder of the paper is organized as follows. In Section 2 we recall the WKB ansatz and the limiting Hamilton–Jacobi/eigenvalue structure, which motivates our AP design. In Section 3 we present the detailed semi-discrete scheme and derive the main stability estimates, including positivity and uniform bounds. Section 4 is devoted to numerical experiments that validate the efficiency of the algorithm and its ability to capture the fittest trait in challenging regimes. Additional details on monotone numerical Hamiltonians and conservative fluxes are provided in the appendix.

2 Preliminary

2.1 WKB ansatz

To efficiently capture the concentration phenomenon in the phenotypic variable, we adopt the classical WKB ansatz

$$n_\varepsilon(x, \theta, t) = W_\varepsilon(x, \theta, t) \exp\left(\frac{u_\varepsilon(\theta, t)}{\varepsilon}\right). \quad (2.2)$$

Formally, as $\varepsilon \rightarrow 0$ and $t \rightarrow +\infty$, this decomposition separates the macroscopic amplitude W_ε from the rapidly varying exponential profile governed by u_ε , which characterizes the concentration dynamics in the phenotypic direction.

As suggested in [?], a formal asymptotic analysis shows that, as $\varepsilon \rightarrow 0$, the steady state $W_\varepsilon(x, \theta)$ converges to a positive eigenfunction $\mathcal{N}(x, \theta)$ solving

$$-D(\theta)\Delta_x \mathcal{N} = \mathcal{N}(K(x) - \rho(\cdot)) + \mathcal{N} \mathcal{H}(\theta, \rho(\cdot)), \quad x \in \Omega, \quad (2.3)$$

with the homogeneous Neumann boundary condition in x -direction and periodic boundary condition in θ -direction, where $H(\theta, \rho(\cdot))$ denotes the associated principal eigenvalue. Moreover, the limiting steady state $u(\theta)$ satisfies the constrained Hamilton–Jacobi equation

$$\begin{cases} |\nabla_\theta u(\theta)|^2 = \mathcal{H}(\theta, \rho(\cdot)), \\ \max_\theta u(\theta) = 0. \end{cases} \quad (2.4)$$

Motivated by this limiting behavior, in order to construct an asymptotic preserving scheme, the effective Hamiltonian $\mathcal{H}$ must be incorporated into the evolution equations for both W_ε and u_ε . Substituting the WKB ansatz (2.2) into equation (??) and assigning all the leading exponential contributions to the phase function u_ε , we obtain

$$\partial_t u_\varepsilon - |\nabla_\theta u_\varepsilon|^2 - \varepsilon \Delta_\theta u_\varepsilon = -\mathcal{H}(\theta, t). \quad (2.5)$$

With u_ε determined by (2.5), the remaining terms yield the evolution equation for the amplitude W_ε , leading to the coupled reformulated system

$$\begin{cases} \partial_t u_\varepsilon - |\nabla_\theta u_\varepsilon|^2 - \varepsilon \Delta_\theta u_\varepsilon = -\mathcal{H}(\theta, t), \\ \varepsilon \partial_t W_\varepsilon - D(\theta) \Delta_x W_\varepsilon - \varepsilon^2 \Delta_\theta W_\varepsilon - 2\varepsilon \nabla_\theta W_\varepsilon \cdot \nabla_\theta u_\varepsilon = W_\varepsilon (K(x) - \rho_\varepsilon(x) + \mathcal{H}(\theta, t)). \end{cases} \quad (2.6)$$

2.2 Rare mutation limit of the steady state

3 Detailed scheme

3.1 Semi-discrete numerical discretization

By considering semi-discrete schemes, we are able to isolate the effects of the spatial and phenotypic discretizations without additional complications introduced by time stepping. The extension to fully discrete schemes, obtained by applying standard time discretization techniques, follows along similar lines and is therefore omitted. Besides, for clarity of presentation, we restrict our analysis to the one-dimensional spatial case. Extensions to higher-dimensional spatial domains are straightforward and hence omitted.

In the spatial direction, we introduce a uniform grid $\{x_j\}_{j=1}^J$, while in the phenotypic direction the variable θ is discretized on grid points $\{\theta_k\}_{k=1}^K$. The numerical approximation of $W_\varepsilon(t, x_j, \theta_k)$

and $u_\varepsilon(t, \theta_k)$ are $W_{j,k}^\varepsilon(t)$ and $u_k^\varepsilon(t)$, respectively. Let Δ_x denote the Laplace operator in the spatial variable. We assume that it is approximated by a discrete operator A_h acting on grid functions defined on $\{x_j\}_{j=1}^J$, and satisfying the stability bound

$$\|A_h v\|_\infty \leq C_h \|v\|_\infty,$$

where $C_h > 0$ depends on the mesh size but is independent of v . A typical example is the standard second-order centered difference operator δ_x^2 . Similarly, we approximate $\partial_{\theta\theta}$ by the centered second-order operator δ_θ^2 , defined by

$$\delta_x^2 W_{j,k} = \frac{W_{j+1,k} - 2W_{j,k} + W_{j-1,k}}{(\Delta x)^2}, \quad \delta_\theta^2 W_{j,k} = \frac{W_{j,k+1} - 2W_{j,k} + W_{j,k-1}}{(\Delta \theta)^2}.$$

In the reformulated WKB system (2.6), the θ -direction contains first-order terms involving both W_ε and u_ε . To facilitate the subsequent positivity and boundedness analysis for W_ε , as well as the viscosity-solution framework for u_ε , we adopt monotone upwind-type discretizations in the θ -direction. We introduce the notation

$$H_1(\partial_\theta u_\varepsilon) := -|\partial_\theta u_\varepsilon|^2, \quad H_2(\partial_\theta W_\varepsilon, \partial_\theta u_\varepsilon) := -2\varepsilon \partial_\theta W_\varepsilon \partial_\theta u_\varepsilon.$$

Instead of discretizing H_1 in its nonconservative product form, we rewrite it using the product rule:

$$-2\varepsilon \partial_\theta W_\varepsilon \partial_\theta u_\varepsilon = \partial_\theta (-2\varepsilon W_\varepsilon \partial_\theta u_\varepsilon) + 2\varepsilon W_\varepsilon \partial_{\theta\theta} u_\varepsilon. \quad (3.7)$$

This reformulation allows us to discretize the first term in (3.7) by a conservative monotone numerical flux, which is crucial for establishing a discrete maximum principle (and hence $W_\varepsilon \geq 0$) under standard time discretizations.

Monotone numerical Hamiltonian for the Hamilton–Jacobi term. For the Hamilton–Jacobi term $H_1(\partial_\theta u_\varepsilon)$, we approximate the nonlinear Hamiltonian $H_1(p) = -|p|^2$ by a monotone numerical Hamiltonian $\hat{H}(a, b)$ depending on the one-sided slopes

$$D_\theta^- u_k = \frac{u_k - u_{k-1}}{\Delta \theta}, \quad D_\theta^+ u_k = \frac{u_{k+1} - u_k}{\Delta \theta}.$$

Specifically, we define

$$H_1(\partial_\theta u) \Big|_{\theta=\theta_k} \approx \hat{H}(D_\theta^- u_k, D_\theta^+ u_k),$$

where the numerical Hamiltonian $\hat{H}$ satisfies the consistency condition $\hat{H}(p, p) = H_1(p)$ and the monotonicity property that it is nondecreasing in its first argument and nonincreasing in its second argument. These conditions ensure that the resulting scheme is monotone and fits into the standard framework of monotone approximations for viscosity solutions of Hamilton–Jacobi equations.

Typical choices of $\hat{H}(a, b)$ include the Godunov and Lax–Friedrichs numerical Hamiltonians. In practice, Roe-type numerical Hamiltonians combined with suitable entropy fixes are often employed to reduce numerical dissipation while preserving monotonicity. Higher-order accuracy can be achieved by applying ENO, WENO, or MUSCL-type reconstructions to the one-sided slopes $D_\theta^- u_k$ and $D_\theta^+ u_k$. A detailed example can be found in Appendix A.

Monotone flux for the conservative coupling term. For the conservative coupling term $\partial_\theta(-W \partial_\theta u)$, we approximate the flux $F(W, p) = -Wp$, where $p = \partial_\theta u$, by a conservative monotone numerical flux $\widehat{F}_{k+\frac{1}{2}}$. Specifically, we set

$$\partial_\theta(-Wp) \Big|_{\theta=\theta_k} \approx \frac{\widehat{F}_{k+\frac{1}{2}} - \widehat{F}_{k-\frac{1}{2}}}{\Delta\theta}, \quad \widehat{F}_{k+\frac{1}{2}} = \widehat{F}(W_{j,k}, W_{j,k+1}; p_{k+\frac{1}{2}}),$$

where the numerical flux $\widehat{F}(W_L, W_R; p)$ is consistent, i.e. $\widehat{F}(W, W; p) = -2\varepsilon Wp$, and monotone with respect to W_L and W_R in the sense that $\partial_{W_L} \widehat{F} \geq 0$ and $\partial_{W_R} \widehat{F} \leq 0$, uniformly in p . The interface value $p_{k+\frac{1}{2}}$ is obtained from the left and right reconstructed slopes at $\theta_{k+\frac{1}{2}}$. In particular, it can be selected via a Godunov-type procedure consistent with the upwind discretization of the coupling flux. Alternatively, an averaged value of the left and right reconstructed slopes may be used, corresponding to a Lax–Friedrichs-type treatment of the coupling flux. A detailed example can be found in Appendix B.

Semi-discrete scheme for W and u . For each $k = 1, \dots, K$, let the discrete density $\rho_j(t)$ be defined by

$$\rho_j^\varepsilon(t) = \Delta\theta \sum_{k=1}^K W_{j,k}^\varepsilon(t) \exp\left(\frac{u_k^\varepsilon(t)}{\varepsilon}\right), \quad j = 1, \dots, J. \quad (3.8)$$

Given $\boldsymbol{\rho}(t) = (\rho_1^\varepsilon(t), \dots, \rho_J^\varepsilon(t))$, we consider the discrete eigenvalue problem associated with (2.3):

$$-D(\theta_k) \delta_x^2 \mathcal{N}_j = \mathcal{N}_j (K(x_j) - \rho_j^\varepsilon(t)) + \mathcal{N}_j \mathcal{H}_k(\boldsymbol{\rho}(t)), \quad j = 1, \dots, J, \quad (3.9)$$

where $\mathcal{H}_k(\boldsymbol{\rho}(t))$ and $\mathcal{N}_j$ denote the numerical approximations of the principal eigenvalue and the corresponding eigenfunction $\mathcal{N}(x_j)$, respectively.

With the above discretizations, the detailed semi-discrete scheme reads as follows. The phase function $u_k^\varepsilon(t)$ is governed by the semi-discrete Hamilton–Jacobi equation

$$\frac{d}{dt} u_k^\varepsilon + \widehat{H}(D_\theta^- u_k^\varepsilon, D_\theta^+ u_k^\varepsilon) - \varepsilon \delta_\theta^2 u_k^\varepsilon = -\mathcal{H}_k(\boldsymbol{\rho}(t)), \quad k = 1, \dots, K, \quad (3.10)$$

where $\mathcal{H}_k(\boldsymbol{\rho}(t))$ denotes the principal eigenvalue associated with the discrete eigenvalue problem (3.9). For $j = 1, \dots, J$ and $k = 1, \dots, K$, the amplitude $W_{j,k}^\varepsilon(t)$ satisfies

$$\begin{aligned} \varepsilon \frac{d}{dt} W_{j,k}^\varepsilon - D(\theta_k) (A_h W^\varepsilon)_{j,k} - \varepsilon^2 \delta_\theta^2 W_{j,k}^\varepsilon + 2\varepsilon \frac{\widehat{F}_{j,k+\frac{1}{2}} - \widehat{F}_{j,k-\frac{1}{2}}}{\Delta\theta} \\ = W_{j,k}^\varepsilon (K(x_j) - \rho_j^\varepsilon(t) + \mathcal{H}_k(\boldsymbol{\rho}(t)) - 2\varepsilon \delta_\theta^2 u_k^\varepsilon). \end{aligned} \quad (3.11)$$

Here the numerical Hamiltonian $\widehat{H}(a, b)$ is consistent and locally Lipschitz, and the numerical flux $\widehat{F}(a)$ is also consistent and locally Lipschitz in the sense that ...

Semi-discrete scheme for $n_{j,k}^\varepsilon(t)$. Multiplying (3.11) by $\exp(u_k^\varepsilon/\varepsilon)$ and (3.10) by $W_{j,k}^\varepsilon \exp(u_k^\varepsilon/\varepsilon)$, and using the fact that A_h acts only on the spatial index j , we arrive at the semi-discrete equation for $n_{j,k}^\varepsilon$:

$$\varepsilon \frac{d}{dt} n_{j,k}^\varepsilon - D(\theta_k) A_h (n_{\cdot,k}^\varepsilon)_j = (K(x_j) - \rho_j^\varepsilon) n_{j,k}^\varepsilon + R_{j,k}^\varepsilon, \quad (3.12)$$

where R_j^ε collects all the terms arising from the θ -direction discretization and takes the form

$$\begin{aligned}
R_{j,k}^\varepsilon &= \varepsilon^2 e^{u_k^\varepsilon(t)/\varepsilon} \delta_\theta^2 W_{j,k}^\varepsilon(t) \\
&\quad - \varepsilon W_{j,k}^\varepsilon(t) e^{u_k^\varepsilon(t)/\varepsilon} \delta_\theta^2 u_k^\varepsilon(t) \\
&\quad - W_{j,k}^\varepsilon(t) e^{u_k^\varepsilon(t)/\varepsilon} \widehat{H}(D_\theta^- u_k^\varepsilon(t), D_\theta^+ u_k^\varepsilon(t)) \\
&\quad - 2\varepsilon e^{u_k^\varepsilon(t)/\varepsilon} \frac{\widehat{F}_{j,k+\frac{1}{2}}(t) - \widehat{F}_{j,k-\frac{1}{2}}(t)}{\Delta\theta}.
\end{aligned} \tag{3.13}$$

Here $R_{j,k}^\varepsilon$ collects all the contributions arising from the θ -direction discretization. In the continuous setting, these terms correspond to the diffusion term $\varepsilon^2 \partial_{\theta\theta} n$. In the present semi-discrete WKB formulation, however, the θ -direction diffusion manifests itself through the combined remainder $R_{j,k}^\varepsilon$ given in (3.13).

3.2 A priori estimates

To establish numerical stability and the asymptotic-preserving property, we derive several *a priori* estimates for the semi-discrete WKB system. Due to the coupled structure of the phase and amplitude equations, the analysis relies on a bootstrap argument: we first establish positivity and basic invariance properties, and then derive discrete regularity estimates for the phase function under a boundedness assumption on the density, which will be verified subsequently.

Step 1: Positivity of the amplitude.

Lemma 3.1 (Positivity). *Assume $W_{j,k}^\varepsilon(0) \geq 0$ for all j, k . Then the semi-discrete system (3.11) preserves nonnegativity, i.e.*

$$W_{j,k}^\varepsilon(t) \geq 0 \quad \forall j, k, t \geq 0.$$

Proof. We rewrite (3.11) as a finite-dimensional ODE system

$$\varepsilon \dot{\mathbf{W}} = \mathbf{L}\mathbf{W} + \mathbf{G}(\mathbf{W}, t),$$

where $\mathbf{L}$ contains the linear diffusion operator in x and the monotone conservative discretization in θ , and $\mathbf{G}(\mathbf{W}, t)$ denotes the reaction term of the form $\Xi_{j,k}(t)W_{j,k}^\varepsilon$.

By construction, $\mathbf{L}$ is a Metzler-type operator with nonnegative off-diagonal entries: the diffusion operator in x is associated with an M -matrix, while the discretization in the θ -direction is monotone (upwind or Godunov type). Hence the linear semiflow generated by $\mathbf{L}$ is order-preserving. Moreover, the nonlinear term $\mathbf{G}$ is quasi-monotone and nonnegative whenever $W_{j,k}^\varepsilon$ touches 0. Therefore, the nonnegative cone is invariant under the induced semiflow, which implies $W_{j,k}^\varepsilon(t) \geq 0$ for all $t \geq 0$. $\square$

Step 2: Discrete slope bound for the phase function.

Lemma 3.2 (Local-in-time discrete bounds for u). *Assume that the initial phase is bounded in the sense that*

$$\max_k u_k^\varepsilon(0) \leq C_0 \varepsilon,$$

discretely Lipschitz, i.e.,

$$\max_k \left(|D_\theta^- u_k^\varepsilon(0)| + |D_\theta^+ u_k^\varepsilon(0)| \right) \leq L_0, \tag{3.14}$$

and that the initial discrete density $\rho_j(0), j = 1, \dots, J$, is bounded in space. Then there exists T , possibly depending on ε , and some constants $C_T, L_T > 0$ depending on T , such that

$$\max_{t \in [0, T]} \max_k u_k^\varepsilon(t) \leq C_T \varepsilon, \quad \max_{t \in [0, T]} \max_k \left(|D_\theta^- u_k^\varepsilon(t)| + |D_\theta^+ u_k^\varepsilon(t)| \right) \leq L_T.$$

Proof. Since the semi-discrete system (3.10)–(3.11) is a finite-dimensional ODE system with locally Lipschitz right-hand side, for each fixed $\varepsilon > 0$ there exists a (possibly small) time $T > 0$ such that the solution $\{u_k^\varepsilon(t), W_{j,k}^\varepsilon(t)\}$ exists and is continuous on $[0, T]$. In particular, $u_k^\varepsilon(t)$ depends continuously on time and on the initial data.

By assumption, the initial phase satisfies $\max_k u_k^\varepsilon(0) \leq C_0 \varepsilon$ and is discretely Lipschitz in the sense of (3.14). Since the one-sided derivatives $D_\theta^\pm u_k^\varepsilon$ are obtained from u^ε by a fixed reconstruction procedure (possibly nonlinear, such as ENO/WENO or MUSCL), which is continuous with respect to the nodal values of u^ε , it follows from continuity that there exists a time $T > 0$, possibly depending on ε , and a constant $L_T > 0$ such that

$$\max_{t \in [0, T]} \max_k \left(|D_\theta^- u_k^\varepsilon(t)| + |D_\theta^+ u_k^\varepsilon(t)| \right) \leq L_T.$$

We next control the amplitude of $u^\varepsilon(t)$. By continuity of the solution and boundedness of the initial discrete density $\rho_j(0)$, the source term $\mathcal{H}_k(\rho(t))$ remains bounded on $[0, T]$, i.e., there exists $C_T > 0$ such that

$$\max_{t \in [0, T]} \max_k |\mathcal{H}_k(\rho(t))| \leq C_T.$$

Evaluating the semi-discrete Hamilton–Jacobi equation (3.10) and using the boundedness of $\widehat{H}(D_\theta^- u_k^\varepsilon, D_\theta^+ u_k^\varepsilon)$ on $[0, T]$, we infer that $\partial_t u_k^\varepsilon(t)$ is bounded on $[0, T]$. Consequently, there exists (possibly after reducing T) a constant $C_T > 0$ such that

$$\max_k u_k^\varepsilon(t) \leq C_T \varepsilon, \quad t \in [0, T].$$

This proves the local-in-time boundedness and discrete Lipschitz estimates stated in the lemma. $\square$

Lemma 3.2 provides only a local-in-time control of the phase function. In the next step, we derive uniform bounds for the density ρ^ε , which allow us to remove the ε -dependence of the existence time and close the bootstrap argument. Similarly, by the continuity argument, we can also get the local-in-time discrete bound for W , the result of which can be fomulated as the following lemma.

Lemma 3.3 (Local-in-time weighted first-difference bound for W). *Assume that $W_{j,k}^\varepsilon(0) \geq 0$, for all j, k , then there exists T such that for all $t \in [0, T]$ and all j ,*

$$\Delta\theta \sum_{k=1}^K \exp(u_k^\varepsilon(t)/\varepsilon) |\delta_\theta^+ W_{j,k}^\varepsilon(t)| \leq M_T \rho_{\varepsilon,j}(t), \quad \delta_\theta^+ W_{j,k}^\varepsilon = \frac{W_{j,k+1}^\varepsilon - W_{j,k}^\varepsilon}{\Delta\theta}, \quad (3.15)$$

where M_T is some constant depending on T .

Proof. We choose the time $T > 0$ such that the local discrete slope bound for the phase function given in Lemma 3.2 holds on $[0, T]$. Let $g_k(t) := \exp(u_k^\varepsilon(t)/\varepsilon)$. By Lemma 3.1 (positivity preservation), the assumption $W_{j,k}^\varepsilon(0) \geq 0$ implies

$$W_{j,k}^\varepsilon(t) \geq 0, \quad \forall j, k, t \geq 0,$$

in particular on $[0, T]$.

Fix $t \in [0, T]$ and j . Since $W_{j,k}^\varepsilon(t) \geq 0$, we have

$$|W_{j,k+1}^\varepsilon(t) - W_{j,k}^\varepsilon(t)| \leq W_{j,k+1}^\varepsilon(t) + W_{j,k}^\varepsilon(t).$$

Therefore,

$$\begin{aligned} \Delta\theta \sum_{k=1}^K g_k(t) |\delta_\theta^+ W_{j,k}^\varepsilon(t)| &= \sum_{k=1}^K g_k(t) |W_{j,k+1}^\varepsilon(t) - W_{j,k}^\varepsilon(t)| \\ &\leq \sum_{k=1}^K g_k(t) W_{j,k+1}^\varepsilon(t) + \sum_{k=1}^K g_k(t) W_{j,k}^\varepsilon(t). \end{aligned}$$

The second sum equals $\rho_{\varepsilon,j}(t)$ by definition. For the first sum, we use a local comparability of the weights. Since u^ε is continuous in time and satisfies a local discrete slope bound on $[0, T]$ (Lemma 3.2), there exists a constant $\Gamma_T > 0$ such that

$$\max_{t \in [0, T]} \max_k \left| \frac{u_{k+1}^\varepsilon(t) - u_k^\varepsilon(t)}{\varepsilon} \right| \leq \Gamma_T. \quad (3.16)$$

Consequently,

$$g_k(t) = e^{u_k^\varepsilon(t)/\varepsilon} \leq e^{\Gamma_T} e^{u_{k+1}^\varepsilon(t)/\varepsilon} = e^{\Gamma_T} g_{k+1}(t), \quad t \in [0, T],$$

and hence

$$\sum_{k=1}^K g_k(t) W_{j,k+1}^\varepsilon(t) \leq e^{\Gamma_T} \sum_{k=1}^K g_{k+1}(t) W_{j,k+1}^\varepsilon(t) = e^{\Gamma_T} \sum_{k=1}^K g_k(t) W_{j,k}^\varepsilon(t) = e^{\Gamma_T} \rho_{\varepsilon,j}(t).$$

Putting the above estimates together yields

$$\Delta\theta \sum_{k=1}^K g_k(t) |\delta_\theta^+ W_{j,k}^\varepsilon(t)| \leq (1 + e^{\Gamma_T}) \rho_{\varepsilon,j}(t), \quad t \in [0, T].$$

This proves (3.15) with $M_T := 1 + e^{\Gamma_T}$. □

Step 3: Local control of the remainder term.

Lemma 3.4 (Consistency of the weighted θ -remainder evaluated at the exact solution). *Let $(u^\varepsilon, W^\varepsilon)$ be the exact (continuous-in- θ) solution pair of the underlying WKB system, and set*

$$n^\varepsilon(x, \theta, t) = W^\varepsilon(x, \theta, t) e^{u^\varepsilon(\theta, t)/\varepsilon}.$$

Let $\{\theta_k\}_{k=1}^K$ be a uniform periodic grid with step size $\Delta\theta$, and define the sampling

$$u_k^\varepsilon(t) := u^\varepsilon(\theta_k, t), \quad W_{j,k}^\varepsilon(t) := W^\varepsilon(x_j, \theta_k, t), \quad n_{j,k}^\varepsilon(t) := n^\varepsilon(x_j, \theta_k, t).$$

Let $D_k := D(\theta_k)$, $\alpha_k := 1/D_k$, and let $R_{j,k}^\varepsilon(t)$ be the discrete θ -remainder defined by (3.13) (using the same numerical Hamiltonian $\hat{H}$ and numerical flux $\hat{F}$ as in the scheme). Define the weighted aggregate

$$\mathcal{R}_j^{\varepsilon, D}(t) := \Delta\theta \sum_{k=1}^K \alpha_k R_{j,k}^\varepsilon(t).$$

Assume $D \in W^{2,\infty}(\mathbb{T})$ with $D \geq D_{\min} > 0$, and assume that for $t \in [0, T]$

$$u^\varepsilon(\cdot, t) \in W^{3,\infty}(\mathbb{T}), \quad W^\varepsilon(x_j, \cdot, t) \in W^{2,\infty}(\mathbb{T}),$$

with bounds possibly depending on ε (but finite for each fixed ε). Assume moreover that $\hat{H}$ and $\hat{F}$ are consistent first-order monotone approximations of their continuous counterparts and Lipschitz in their arguments.

Then for each fixed $\varepsilon > 0$ there exists a constant $C_{\text{cons}}(\varepsilon) > 0$ (depending on the above Sobolev bounds of the exact solution and on $\|(1/D)_{\theta\theta}\|_{L^\infty}$, but independent of $\Delta\theta$) such that for all $t \in [0, T]$,

$$\left| \mathcal{R}_j^{\varepsilon,D}(t) - \varepsilon^2 \Delta\theta \sum_{k=1}^K n_{j,k}^\varepsilon(t) \delta_\theta^2 \left(\frac{1}{D_k} \right) \right| \leq C_{\text{cons}}(\varepsilon) \Delta\theta. \quad (3.17)$$

In particular, the leading term on the left-hand side depends explicitly on $\delta_\theta^2(1/D)$ and vanishes when D is constant.

Proof. Fix j and t . Since $n^\varepsilon = W^\varepsilon e^{u^\varepsilon/\varepsilon}$ is smooth in θ , the central difference δ_θ^2 is consistent:

$$\delta_\theta^2 n_{j,k}^\varepsilon(t) = \partial_{\theta\theta} n^\varepsilon(x_j, \theta_k, t) + \mathcal{O}(\Delta\theta^2),$$

and similarly $D_\theta^\pm u_k^\varepsilon = \partial_\theta u^\varepsilon(\theta_k, t) + \mathcal{O}(\Delta\theta)$. By the consistency and Lipschitz continuity of $\hat{H}$ and $\hat{F}$, we have

$$\hat{H}(D_\theta^- u_k^\varepsilon, D_\theta^+ u_k^\varepsilon) = (\partial_\theta u^\varepsilon)^2(\theta_k, t) + \mathcal{O}(\Delta\theta), \quad \frac{\hat{F}_{j,k+\frac{1}{2}} - \hat{F}_{j,k-\frac{1}{2}}}{\Delta\theta} = \partial_\theta(\text{cont. flux})(x_j, \theta_k, t) + \mathcal{O}(\Delta\theta),$$

where the $\mathcal{O}(\Delta\theta)$ constants depend on $\|u^\varepsilon(\cdot, t)\|_{W^{3,\infty}}$ and the Lipschitz constants of $\hat{H}, \hat{F}$. Inserting the exact solution samples into the discrete definition (3.13) and comparing with the continuous identity that the θ -diffusion term equals $\varepsilon^2 \partial_{\theta\theta} n^\varepsilon$, we obtain a pointwise consistency relation of the form

$$R_{j,k}^\varepsilon(t) = \varepsilon^2 \delta_\theta^2 n_{j,k}^\varepsilon(t) + \tau_{j,k}^\varepsilon(t), \quad |\tau_{j,k}^\varepsilon(t)| \leq C_{\text{cons}}(\varepsilon) \Delta\theta,$$

where $C_{\text{cons}}(\varepsilon)$ depends on the stated Sobolev bounds and on the numerical Lipschitz constants, but not on $\Delta\theta$. Multiplying by α_k and summing over k yields

$$\mathcal{R}_j^{\varepsilon,D}(t) = \varepsilon^2 \Delta\theta \sum_{k=1}^K \alpha_k \delta_\theta^2 n_{j,k}^\varepsilon(t) + \Delta\theta \sum_{k=1}^K \alpha_k \tau_{j,k}^\varepsilon(t).$$

By periodic summation by parts,

$$\Delta\theta \sum_{k=1}^K \alpha_k \delta_\theta^2 n_{j,k}^\varepsilon = \Delta\theta \sum_{k=1}^K n_{j,k}^\varepsilon \delta_\theta^2 \alpha_k = \Delta\theta \sum_{k=1}^K n_{j,k}^\varepsilon \delta_\theta^2 \left(\frac{1}{D_k} \right),$$

which gives the leading term in (3.17). Finally, since $\alpha_k \leq 1/D_{\min}$, we have

$$\left| \Delta\theta \sum_{k=1}^K \alpha_k \tau_{j,k}^\varepsilon(t) \right| \leq \frac{1}{D_{\min}} \Delta\theta \sum_{k=1}^K |\tau_{j,k}^\varepsilon(t)| \leq C_{\text{cons}}(\varepsilon) \Delta\theta,$$

which completes the proof. $\square$

Lemma 3.5 (Weighted remainder estimate after division by $D(\theta)$). *Let $\{\theta_k\}_{k=1}^K$ be a uniform periodic grid on $\mathbb{T}$ with step size $\Delta\theta$, and define $D_k := D(\theta_k)$. Assume that there exists $D_{\min} > 0$ such that*

$$D_k \geq D_{\min} \quad \forall k = 1, \dots, K. \quad (3.18)$$

For each fixed j and t , define

$$E_k := e^{u_k^\varepsilon(t)/\varepsilon}, \quad A_k := W_{j,k}^\varepsilon(t) E_k, \quad \alpha_k := \frac{1}{D_k}.$$

Assume that the discrete operators $D_\theta^\pm$ and δ_θ^2 are defined by

$$D_\theta^+ a_k = \frac{a_{k+1} - a_k}{\Delta\theta}, \quad D_\theta^- a_k = \frac{a_k - a_{k-1}}{\Delta\theta}, \quad \delta_\theta^2 a_k = \frac{a_{k+1} - 2a_k + a_{k-1}}{\Delta\theta^2},$$

with periodic indexing $a_{k+K} = a_k$. For each j , introduce the D -weighted aggregate remainder

$$\mathcal{R}_j^{\varepsilon,D}(t) := \Delta\theta \sum_{k=1}^K \frac{R_{j,k}^\varepsilon(t)}{D_k}.$$

Assume that, locally, there exists $C_ > 0$ independent of ε and $\Delta\theta$ such that*

$$\|D_\theta^\pm u^\varepsilon(\cdot, t)\|_{\ell^\infty} \leq C_*, \quad \|W_{j,\cdot}^\varepsilon(t)\|_{\ell^\infty} \leq C_*, \quad \Delta\theta \sum_{k=1}^K W_{j,k}^\varepsilon e^{\frac{u_k^\varepsilon(t)}{\varepsilon}} \leq C_*. \quad (3.19)$$

There exists a sequence $\{\mathcal{T}_{j,k}^\varepsilon(t)\}_{k=1}^K$ such that, for all k ,

$$R_{j,k}^\varepsilon(t) = \varepsilon^2 \delta_\theta^2 A_k(t) + \mathcal{T}_{j,k}^\varepsilon(t), \quad (3.20)$$

and the truncation error satisfies the weighted aggregate bound

$$\left| \Delta\theta \sum_{k=1}^K \alpha_k \mathcal{T}_{j,k}^\varepsilon(t) \right| \leq C \Delta\theta \left(\varepsilon \|D_\theta^+ W_{j,\cdot}^\varepsilon(t)\|_{\ell^1} + \|W_{j,\cdot}^\varepsilon(t)\|_{\ell^1} \right), \quad t \in [0, T_*], \quad (3.21)$$

with C independent of $\varepsilon, \Delta\theta$.

Then there exists a constant $C > 0$, independent of ε and $\Delta\theta$, such that for all $t \in [0, T_]$ and each fixed j ,*

$$|\mathcal{R}_j^{\varepsilon,D}(t)| \leq C_{D,2} \varepsilon^2 \rho_j^\varepsilon(t) + C \Delta\theta \left(\varepsilon \|D_\theta^+ W_{j,\cdot}^\varepsilon(t)\|_{\ell^1} + \|W_{j,\cdot}^\varepsilon(t)\|_{\ell^1} \right), \quad (3.22)$$

where $\rho_j^\varepsilon(t) := \Delta\theta \sum_{k=1}^K n_{j,k}^\varepsilon(t) = \Delta\theta \sum_{k=1}^K A_k$. In particular, the first term on the right-hand side depends explicitly on the discrete second derivative of $1/D$ (through $C_{D,2}$), and it vanishes when D is constant (since then $\delta_\theta^2 \alpha \equiv 0$).

Proof. Fix j and $t \in [0, T_*]$. By the reduction (3.20),

$$\mathcal{R}_j^{\varepsilon,D}(t) = \varepsilon^2 \Delta\theta \sum_{k=1}^K \alpha_k \delta_\theta^2 A_k + \Delta\theta \sum_{k=1}^K \alpha_k \mathcal{T}_{j,k}^\varepsilon.$$

We estimate these two terms.

Step 1: the non-vanishing ε^2 contribution. Applying SBP (A1) twice (equivalently, using symmetry of δ_θ^2 on periodic grids), we obtain

$$\Delta\theta \sum_{k=1}^K \alpha_k \delta_\theta^2 A_k = \Delta\theta \sum_{k=1}^K A_k \delta_\theta^2 \alpha_k.$$

Therefore,

$$\varepsilon^2 \left| \Delta\theta \sum_{k=1}^K \alpha_k \delta_\theta^2 A_k \right| = \varepsilon^2 \left| \Delta\theta \sum_{k=1}^K A_k \delta_\theta^2 \alpha_k \right| \leq \varepsilon^2 \|\delta_\theta^2 \alpha\|_{\ell^\infty} \Delta\theta \sum_{k=1}^K A_k \leq C_{D,2} \varepsilon^2 \rho_j^\varepsilon(t).$$

If D is constant, then α is constant and $\delta_\theta^2 \alpha \equiv 0$, so this term vanishes.

Step 2: the $\Delta\theta$ -small truncation error. This is exactly (3.21).

Combining the above two bounds yields (??). $\square$

Theorem 3.1 (Uniform boundedness of ρ). *Assume that $D(\theta)$ is positive and bounded away from 0, i.e.,*

$$0 < D_{\min} \leq D(\theta) \leq D_{\max} < \infty,$$

and that D is sufficiently regular so that the discrete second difference $\delta_\theta^2(1/D_k)$ is uniformly bounded. Then there exists a constant $C > 0$, independent of ε , such that

$$0 \leq \rho_j^\varepsilon(t) \leq C, \quad \forall j = 1, \dots, J, \quad t \geq 0.$$

Proof. The proof follows the continuous strategy of dividing the n -equation by $D(\theta)$ before integrating in θ , and avoids any mesh-dependent constant in space.

Step 1. Equation for $n_{j,k}^\varepsilon$ and discrete θ -summation-by-parts. Define the discrete density

$$n_{j,k}^\varepsilon(t) := W_{j,k}^\varepsilon(t) e^{u_k^\varepsilon(t)/\varepsilon}, \quad \rho_j^\varepsilon(t) = \Delta\theta \sum_{k=1}^K n_{j,k}^\varepsilon(t).$$

By construction and Lemma 3.1, we have $n_{j,k}^\varepsilon \geq 0$ and $\rho_j^\varepsilon \geq 0$.

We start from the semi-discrete scheme (3.12). We shall use the discrete summation-by-parts identity (periodic in θ):

$$\Delta\theta \sum_{k=1}^K \phi_k \delta_\theta^2 \psi_k = \Delta\theta \sum_{k=1}^K \psi_k \delta_\theta^2 \phi_k, \quad \text{for any periodic sequences } (\phi_k), (\psi_k). \quad (3.23)$$

Step 2. Divide by D_k and sum in θ : a new ρ -equation. Divide (??) by D_k and sum over k with weight $\Delta\theta$. Introduce the weighted density

$$\tilde{\rho}_j^\varepsilon(t) := \Delta\theta \sum_{k=1}^K \frac{n_{j,k}^\varepsilon(t)}{D_k}.$$

Since $D_{\min} \leq D_k \leq D_{\max}$, we have the pointwise comparison

$$\frac{1}{D_{\max}} \rho_j^\varepsilon \leq \tilde{\rho}_j^\varepsilon \leq \frac{1}{D_{\min}} \rho_j^\varepsilon. \quad (3.24)$$

Summing the spatial term is immediate:

$$\Delta\theta \sum_{k=1}^K \frac{1}{D_k} D_k (A_h n_{\cdot,k}^\varepsilon)_j = (A_h \rho^\varepsilon)_j.$$

For the θ -diffusion term, apply (3.23) with $\phi_k = 1/D_k$ and $\psi_k = n_{j,k}^\varepsilon$ to obtain

$$\Delta\theta \sum_{k=1}^K \frac{1}{D_k} \delta_\theta^2 n_{j,k}^\varepsilon = \Delta\theta \sum_{k=1}^K n_{j,k}^\varepsilon \delta_\theta^2 \left(\frac{1}{D_k} \right).$$

Hence we arrive at

$$\varepsilon \frac{d}{dt} \tilde{\rho}_j^\varepsilon = (A_h \rho^\varepsilon)_j + \varepsilon^2 \Delta\theta \sum_{k=1}^K n_{j,k}^\varepsilon \delta_\theta^2 \left(\frac{1}{D_k} \right) + (K_j - \rho_j^\varepsilon) \tilde{\rho}_j^\varepsilon + \tilde{\mathcal{R}}_{\varepsilon,j}, \quad (3.25)$$

where

$$\tilde{\mathcal{R}}_{\varepsilon,j} := \Delta\theta \sum_{k=1}^K \frac{\mathcal{R}_{j,k}^\varepsilon}{D_k}.$$

Step 3. Bounding the extra terms created by dividing by D_k . Since $\delta_\theta^2(1/D_k)$ is uniformly bounded and $n_{j,k}^\varepsilon \geq 0$, there exists $C_D > 0$ such that

$$\left| \Delta\theta \sum_{k=1}^K n_{j,k}^\varepsilon \delta_\theta^2 \left(\frac{1}{D_k} \right) \right| \leq C_D \Delta\theta \sum_{k=1}^K n_{j,k}^\varepsilon = C_D \rho_j^\varepsilon. \quad (3.26)$$

Moreover, on any finite interval $[0, T]$, Lemma ?? implies

$$|\tilde{\mathcal{R}}_{\varepsilon,j}(t)| \leq \frac{1}{D_{\min}} \Delta\theta \sum_{k=1}^K |\mathcal{R}_{j,k}^\varepsilon(t)| \leq C_T (\Delta\theta^q + \varepsilon \Delta\theta^2) \rho_j^\varepsilon(t), \quad t \in [0, T], \quad (3.27)$$

for some $C_T > 0$ independent of ε .

Plugging (3.26)–(3.27) into (3.25), we obtain the *structural inequality*

$$(A_h \rho^\varepsilon)_j - \rho_j^\varepsilon \tilde{\rho}_j^\varepsilon \geq -K_j \tilde{\rho}_j^\varepsilon - C_T^* \rho_j^\varepsilon - \varepsilon \left| \frac{d}{dt} \tilde{\rho}_j^\varepsilon \right|, \quad t \in [0, T], \quad (3.28)$$

where C_T^* depends on T , D and the remainder bound, but not on ε .

Step 4. Maximum principle in space and a quadratic bound. Fix $T > 0$ and $t \in [0, T]$. Let j^* be such that

$$\rho_{j^*}^\varepsilon(t) = \max_{1 \leq j \leq J} \rho_j^\varepsilon(t) =: M(t).$$

For the standard centered discretization with homogeneous Neumann boundary conditions, the discrete maximum principle yields

$$(A_h \rho^\varepsilon)_{j^*}(t) \leq 0. \quad (3.29)$$

Evaluating (3.28) at $j = j^*$ and using (3.29) gives

$$\rho_{j^*}^\varepsilon(t) \tilde{\rho}_{j^*}^\varepsilon(t) \leq K_{\max} \tilde{\rho}_{j^*}^\varepsilon(t) + C_T^* \rho_{j^*}^\varepsilon(t) + \varepsilon \left| \frac{d}{dt} \tilde{\rho}_{j^*}^\varepsilon(t) \right|. \quad (3.30)$$

It remains to control the last term. From (3.25), using $(A_h \rho^\varepsilon)_{j^*} \leq 0$, (3.26), and (3.27), we infer

$$\varepsilon \frac{d}{dt} \tilde{\rho}_{j^*}^\varepsilon \leq C_T^* \rho_{j^*}^\varepsilon + K_{\max} \tilde{\rho}_{j^*}^\varepsilon,$$

hence

$$\varepsilon \left| \frac{d}{dt} \tilde{\rho}_{j^*}^\varepsilon \right| \leq C_T^* \rho_{j^*}^\varepsilon + K_{\max} \tilde{\rho}_{j^*}^\varepsilon. \quad (3.31)$$

Plugging (3.31) into (3.30) yields

$$\rho_{j^*}^\varepsilon \tilde{\rho}_{j^*}^\varepsilon \leq 2K_{\max} \tilde{\rho}_{j^*}^\varepsilon + 2C_T^* \rho_{j^*}^\varepsilon.$$

Using the comparison (3.24), namely $\tilde{\rho}_{j^*}^\varepsilon \geq \rho_{j^*}^\varepsilon / D_{\max}$ and $\tilde{\rho}_{j^*}^\varepsilon \leq \rho_{j^*}^\varepsilon / D_{\min}$, we obtain a quadratic inequality for $M(t) = \rho_{j^*}^\varepsilon(t)$:

$$\frac{1}{D_{\max}} M(t)^2 \leq \frac{2K_{\max}}{D_{\min}} M(t) + 2C_T^* M(t).$$

Therefore,

$$M(t) \leq D_{\max} \left(\frac{2K_{\max}}{D_{\min}} + 2C_T^* \right) =: C_T, \quad \forall t \in [0, T].$$

Step 5. Global-in-time bound. Since $T > 0$ is arbitrary, the bound extends to all $t \geq 0$. Together with $\rho_j^\varepsilon \geq 0$, we conclude that there exists $C > 0$, independent of ε , such that

$$0 \leq \rho_j^\varepsilon(t) \leq C, \quad \forall j, t \geq 0.$$

□

Theorem 3.2 (Discrete bounds for u and weighted θ -variation of W). *For any fixed $T > 0$. Assume that*

- *the initial phase satisfies*

$$\max_k u_k^\varepsilon(0) \leq C_0 \varepsilon, \quad \max_k \left(|D_\theta^- u_k^\varepsilon(0)| + |D_\theta^+ u_k^\varepsilon(0)| \right) \leq L_0,$$

- *the initial discrete density $\rho_j^\varepsilon(0)$ is bounded in space,*
- *$D(\theta)$ is Lipschitz, and the principal eigenvalue $\mathcal{H}_k(\boldsymbol{\rho})$ depends Lipschitz-continuously on θ_k for $\boldsymbol{\rho}$ in bounded sets.*

Then there exist constants $C_T > 0$, $L_T > 0$ and $M_T > 0$, independent of ε , such that

$$\max_{t \in [0, T]} \max_k u_k^\varepsilon(t) \leq C_T \varepsilon, \quad \max_{t \in [0, T]} \max_k \left(|D_\theta^- u_k^\varepsilon(t)| + |D_\theta^+ u_k^\varepsilon(t)| \right) \leq L_T, \quad (3.32)$$

and, moreover, for all $t \in [0, T]$ and all j ,

$$\Delta \theta \sum_{k=1}^K \exp(u_k^\varepsilon(t)/\varepsilon) |\delta_\theta^+ W_{j,k}^\varepsilon(t)| \leq M_T \rho_{\varepsilon,j}(t), \quad \delta_\theta^+ W_{j,k}^\varepsilon = \frac{W_{j,k+1}^\varepsilon - W_{j,k}^\varepsilon}{\Delta \theta}. \quad (3.33)$$

Proof. We emphasize the bootstrap/continuation structure.

Step 1. Uniform bound for ρ^ε on $[0, T]$. By Theorem (*Uniform boundedness of ρ*), there exists a constant $\bar{\rho}_T > 0$, independent of ε , such that

$$0 \leq \rho_j^\varepsilon(t) \leq \bar{\rho}_T, \quad \forall j = 1, \dots, J, \quad t \in [0, T]. \quad (3.34)$$

In particular, $\rho^\varepsilon(t)$ stays in a bounded set for $t \in [0, T]$.

Step 2. A local bootstrap interval for $(u^\varepsilon, W^\varepsilon)$. Lemma 3.2 (local-in-time discrete bounds for u) applies under the assumptions on the initial phase and the boundedness of $\rho^\varepsilon(0)$, and provides a time $t_1 > 0$ (possibly depending on ε a priori) and constants C_{t_1}, L_{t_1} such that on $[0, t_1]$,

$$\max_{t \in [0, t_1]} \max_k u_k^\varepsilon(t) \leq C_{t_1} \varepsilon, \quad \max_{t \in [0, t_1]} \max_k \left(|D_\theta^- u_k^\varepsilon(t)| + |D_\theta^+ u_k^\varepsilon(t)| \right) \leq L_{t_1}.$$

Moreover, Lemma 3.3 yields (possibly on a smaller time subinterval) the weighted first-difference estimate for W^ε :

$$\Delta\theta \sum_{k=1}^K \exp(u_k^\varepsilon(t)/\varepsilon) |\delta_\theta^+ W_{j,k}^\varepsilon(t)| \leq M_{t_1} \rho_{\varepsilon,j}(t), \quad t \in [0, t_1], \quad \forall j.$$

Step 3. Uniform (in ε) control of the coefficients on bounded sets. By (3.34), the family $\rho^\varepsilon(t)$ remains in a bounded set for $t \in [0, T]$. By the assumed Lipschitz dependence of $\mathcal{H}_k(\rho)$ on θ_k on bounded sets of ρ , there exists a constant $C_T^{\mathcal{H}} > 0$, independent of ε , such that the source term satisfies the discrete Lipschitz estimate (cf. (??))

$$|\mathcal{H}_{k+1}(\rho^\varepsilon(t)) - \mathcal{H}_k(\rho^\varepsilon(t))| \leq C_T^{\mathcal{H}} \Delta\theta, \quad \forall t \in [0, T]. \quad (3.35)$$

Together with the Lipschitz regularity of $D(\theta)$, all coefficients appearing in the semi-discrete u^ε -equation are uniformly controlled on $[0, T]$, with constants independent of ε .

Step 4. Continuation/closure up to time T . Define the maximal time of validity of the bounds by

$$T^* := \sup \left\{ \tau \in (0, T] : \max_{t \in [0, \tau]} \max_k u_k^\varepsilon(t) \leq C \varepsilon, \quad \max_{t \in [0, \tau]} \max_k (|D_\theta^- u_k^\varepsilon(t)| + |D_\theta^+ u_k^\varepsilon(t)|) \leq L \right\},$$

where $C, L > 0$ are constants to be chosen later (independent of ε). By Step 2, we have $T^* > 0$.

Assume by contradiction that $T^* < T$. Since the bounds hold on $[0, T^*]$, the values at time T^* satisfy the same type of initial constraints as at time 0:

$$\max_k u_k^\varepsilon(T^*) \leq C \varepsilon, \quad \max_k \left(|D_\theta^- u_k^\varepsilon(T^*)| + |D_\theta^+ u_k^\varepsilon(T^*)| \right) \leq L,$$

and $\rho^\varepsilon(T^*)$ is bounded by (3.34). Therefore Lemma 3.2 can be re-applied with initial time T^* (in place of 0). Using Step 3, the constants in the slope estimate depend only on T (through $\bar{\rho}_T$ and the Lipschitz constants), but not on ε . Hence there exists a time increment $\delta_T > 0$, depending only on the data and T (and *not* on ε), such that the bounds improve/propagate on $[T^*, T^* + \delta_T]$. This contradicts the maximality of T^* . Consequently $T^* = T$, and we obtain (3.32) on $[0, T]$ with constants C_T, L_T independent of ε .

Step 5. Weighted first-difference bound for W^ε on $[0, T]$. The same continuation argument applies to Lemma 3.3: since ρ^ε remains uniformly bounded on $[0, T]$ and u^ε satisfies (3.32), the

constant in the weighted estimate depends only on T (through the bounds on the coefficients), and thus the estimate can be extended step by step up to time T . This yields (3.33) with a constant M_T independent of ε .

This completes the proof. $\square$

Remark 3.1 (No spurious $\Delta\theta^{-1}$ in the bounds). *The estimate (3.33) is a weighted BV-type bound in the θ -direction and does not introduce any artificial factor $1/\Delta\theta$. Likewise, the phase bound in (3.32) is of the form $\max_k u_k^\varepsilon(t) \lesssim \varepsilon$, which is consistent with the WKB scaling and prevents exponential weights $\exp(u^\varepsilon/\varepsilon)$ from blowing up on finite time intervals.*

3.3 A-P

Theorem 3.3. *Asymptotic preserving property and trait consistency*

Theorem 3.4. $|\theta_{h,\varepsilon=0} - \theta_{\varepsilon=0}| = O(h)$

Theorem 3.5. *Long time evolution towards steady state*

4 Numerical experiments

Acknowledgements X. Ruan acknowledges support from the National Natural Science Foundation of China (Grant number 12201436) and the R&D Program of Beijing Municipal Education Commission from China, grant number KM202310028016. W. Huang acknowledges support from .

A Typical choices of the monotone numerical Hamiltonians

B Typical choices of the monotone numerical fluxes

A representative example is the Godunov (upwind) flux for the linear advection flux $F(W) = aW$ with $a = -2\varepsilon p_{k+\frac{1}{2}}$,

$$\hat{F}_{k+\frac{1}{2}} = 2\varepsilon \left(p_{k+\frac{1}{2}}^- W_{j,k} - p_{k+\frac{1}{2}}^+ W_{j,k+1} \right), \quad p^\pm = \max(\pm p, 0).$$

Example 1: Upwind (Godunov) flux. Let $a_{k+\frac{1}{2}} := -2\varepsilon p_{k+\frac{1}{2}}$. Define

$$\hat{F}_{k+\frac{1}{2}} = \hat{F}^{\text{up}}(W_{j,k}, W_{j,k+1}; p_{k+\frac{1}{2}}) := a_{k+\frac{1}{2}}^+ W_{j,k} + a_{k+\frac{1}{2}}^- W_{j,k+1}, \quad (2.36)$$

where $a^+ := \max(a, 0)$ and $a^- := \min(a, 0)$. Then $\hat{F}^{\text{up}}(W, W; p) = aW = -2\varepsilon Wp$ (consistency), and

$$\partial_{W_L} \hat{F}^{\text{up}} = a^+ \geq 0, \quad \partial_{W_R} \hat{F}^{\text{up}} = a^- \leq 0,$$

so $\hat{F}^{\text{up}}$ is monotone in (W_L, W_R) uniformly in p .

Example 2: Lax–Friedrichs (Rusanov) flux. Let $a_{k+\frac{1}{2}} := -2\varepsilon p_{k+\frac{1}{2}}$ and choose $\alpha_{k+\frac{1}{2}} \geq |a_{k+\frac{1}{2}}|$ (e.g. $\alpha_{k+\frac{1}{2}} = |a_{k+\frac{1}{2}}|$). Define

$$\hat{F}_{k+\frac{1}{2}} = \hat{F}^{\text{LF}}(W_{j,k}, W_{j,k+1}; p_{k+\frac{1}{2}}) := \frac{1}{2}a_{k+\frac{1}{2}}(W_{j,k} + W_{j,k+1}) - \frac{1}{2}\alpha_{k+\frac{1}{2}}(W_{j,k+1} - W_{j,k}). \quad (2.37)$$

Then $\hat{F}^{\text{LF}}(W, W; p) = aW = -2\varepsilon Wp$ (consistency), and

$$\partial_{W_L} \hat{F}^{\text{LF}} = \frac{1}{2}(a + \alpha) \geq 0, \quad \partial_{W_R} \hat{F}^{\text{LF}} = \frac{1}{2}(a - \alpha) \leq 0,$$

hence $\hat{F}^{\text{LF}}$ is monotone provided $\alpha \geq |a|$.

C Discretization

In this appendix, we describe the fully discrete scheme for the system. Assume that (u^n, W^n) are known at time t_n . The update from t_n to t_{n+1} is performed through the following steps.

The density is defined by

$$\rho_\varepsilon(x, t) = \int_0^1 n_\varepsilon(x, \theta, t) d\theta = \int_0^1 W_\varepsilon(x, \theta, t) e^{-u_\varepsilon(\theta, t)/\varepsilon} d\theta.$$

Since $\varepsilon \ll 1$, we approximate the integral using Laplace's method. Let $\theta^*(t)$ be a global maximum point of $u_\varepsilon(\theta, t)$ in $(0, 1)$. Then

$$\rho_\varepsilon(x, t) = \int_0^1 W_\varepsilon(x, \theta, t) e^{u_\varepsilon(\theta, t)/\varepsilon} d\theta \approx W_\varepsilon(x, \theta^*, t) e^{u_\varepsilon(\theta^*, t)/\varepsilon} \sqrt{\frac{2\pi\varepsilon}{\partial_{\theta\theta} u_\varepsilon(\theta^*, t)}}. \quad (3.38)$$

At the discrete spatial nodes x_j ,

$$\rho_j = \rho_\varepsilon(x_j, t) \approx W_\varepsilon(x_j, \theta^*, t) e^{u_\varepsilon(\theta^*, t)/\varepsilon} \sqrt{\frac{2\pi\varepsilon}{\partial_{\theta\theta} u_\varepsilon(\theta^*, t)}}. \quad (3.39)$$

Here θ^* is obtained by performing a local quadratic interpolation of $u_\varepsilon(\theta, t)$ near its discrete maximum.

For each discrete trait value θ_k , the eigenvalue $H_k = H(x, \theta_k, t)$ is computed from the discrete Sturm-Liouville problem,

$$-D(\theta)\delta_x^2 W_{j,k}^n - W_{j,k}(K_j - \rho_j) = W_{j,k} H_k^n. \quad (3.40)$$

The discrete Neumann boundary conditions are imposed as $W_{0,k} = W_{1,k}, W_{N,k} = W_{N-1,k}$.

Let $u_k^n \approx u_\varepsilon(\theta_k, t_n)$, the semi-implicit scheme reads

$$\frac{u_k^{n+1} - u_k^n}{\tau} + \hat{H}(D_\theta^- u_k, D_\theta^+ u_k) - \varepsilon \delta_\theta^2 u_k^{n+1} = -H_k^n. \quad (3.41)$$

Finally, the equation for $W_{j,k}^n \approx W(x_j, \theta_k, t_n)$ is discretized as

$$\begin{aligned} \varepsilon \frac{W_{j,k}^{n+1} - W_{j,k}^n}{\tau} - D_k \delta_x^2 W_{j,k}^{n+1} - \varepsilon^2 \delta_\theta^2 W_{j,k}^{n+1} + 2\varepsilon \frac{\hat{F}_{k+\frac{1}{2}} - \hat{F}_{k-\frac{1}{2}}}{\Delta\theta} + 2\varepsilon W_{j,k} \delta_\theta^2 u_k^{n+1} \\ = W_{j,k}^{n+1}(K_j - \rho_j^n) + W_{j,k}^{n+1} H_k^n. \end{aligned} \quad (3.42)$$

D The discretization of the original problem

$$\begin{aligned}\varepsilon \partial_t n_\varepsilon - D(\theta) \Delta_{xx} n_\varepsilon - \varepsilon^2 \Delta_\theta n_\varepsilon &= n_\varepsilon (K(x) - \rho_\varepsilon(x)), \\ \rho_\varepsilon(x) &= \int_0^1 n_\varepsilon(x, \theta) d\theta\end{aligned}\tag{4.43}$$